Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/16/2006
PI Name:	Santhanakrishnan, Sivakumar Ph.D.		
Project Title:	Oxygen and Fuel Diffusion Flame Studies in Microgravity Motivated by Spacecraft Oxygen Storage Fire Safety		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gore, Jay (Purdue University) Sunderland, Peter (University of Maryland, College Park)		
Grant/Contract No.:	NNC04AA08A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	While past studies and current tests for fire safety have focused on normal diffusion flames (NDFs), the inverse diffusion flame (IDF) configuration is related to possible fire scenarios in spacecraft and extra -vehicular activity (EVA) environments. IDFs also are a potential fire-safety concern for medical care involving enhanced oxygen. In industrial furnaces using oxygenenhanced combustion, flame shape control is important for obtaining a desired heat flux and temperature profile. Flame radiation predictions are functions of the flame volume and surface area. The main objectives of the study, which focuses on NDF and IDF shapes for oxidizer varying from 21% to 100% O2, are:		

	1. to examine the performance of shape predictions of NDFs and IDFs using the extended Roper model with experimental results by varying assumed velocities and temperatures as suitable for 1-g and 0-g. 2. to establish the limits of applicability of these analytical predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is of relevance to fire safety in terrestrial environments where oxygen enhanced combustion is used and in emergency rooms in hospitals etc. where oxygen storage and use is common.
Task Progress:	Comparisons between steady-state computational results and experimental data showed some of the following major findings: 1. Oxygen-enhanced conditions resulted in increased flame temperatures and increased buoyancy effects for normal gravity flames of constant length. 2. The present experiments, analysis, and computations indicate that gravity has a relatively small effect on flame temperature, flame width and flame length of IDFs due to the dominance of convection over buoyancy. This indicates that our planned experiments in normal-gravity IDFs will be highly relevant to oxygen-related spacecraft fire safety. 3. Calculations reveal that buoyant acceleration plays the dominant role for earth-gravity NDFs and that centerline axial velocities can be approximated assuming constant acceleration. 4. Comparisons of maximum temperatures based on present thermo -physical properties and global kinetic rates with adiabatic (2000 species) equilibrium temperatures show that the computations may be overestimating the temperature by underestimating the rates of dissociation reactions. Temperature measurements and detailed chemistry and thermo -physical property calculations are necessary to address this issue.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
Abstracts for Journals and Proceedings	Krishnan SS, Abshire JM, Yuan Z-G, Sunderland PB, Gore JP. "Comparisons of model predictions with measurements of microgravity laminar diffusion flame shapes." 28th Annual Dayton-Cincinnati Aerospace Science Symposium, Dayton, OH, March 2003. Proceedings, 28th Annual Dayton-Cincinnati Aerospace Science Symposium, Dayton, OH, March 2003. , Mar-2003
Awards	Abshire JM. "Graduate Presentation Award - First Place. Presented to Jason M. Abshire (IUPUI graduate student) at the AIAA 2003 Annual Region III Student Conference for his presentation of “Comparisons of Model Predictions with Measurements of Microgravity Laminar Diffusion Flame Shapes." Jan-2004
Dissertations and Theses	Abshire JM. "Laminar Diffusion Flame Shapes under Earth-Gravity and Microgravity Conditions." Thesis, Indiana University- Purdue University Indianapolis, August 2004. , Aug-2004
NASA Technical Documents	Sunderland PB, Yuan Z-G, Krishnan SS, Abshire JM, Gore JP. "Oxygen and Fuel Jet Diffusion Flame Studies in Microgravity Motivated by Spacecraft Oxygen Storage Fire Safety." In: Seventh International Microgravity Combustion Workshop, Cleveland, NASA CP-2003-212376, 2003. pp. 373-376. , Dec-2003